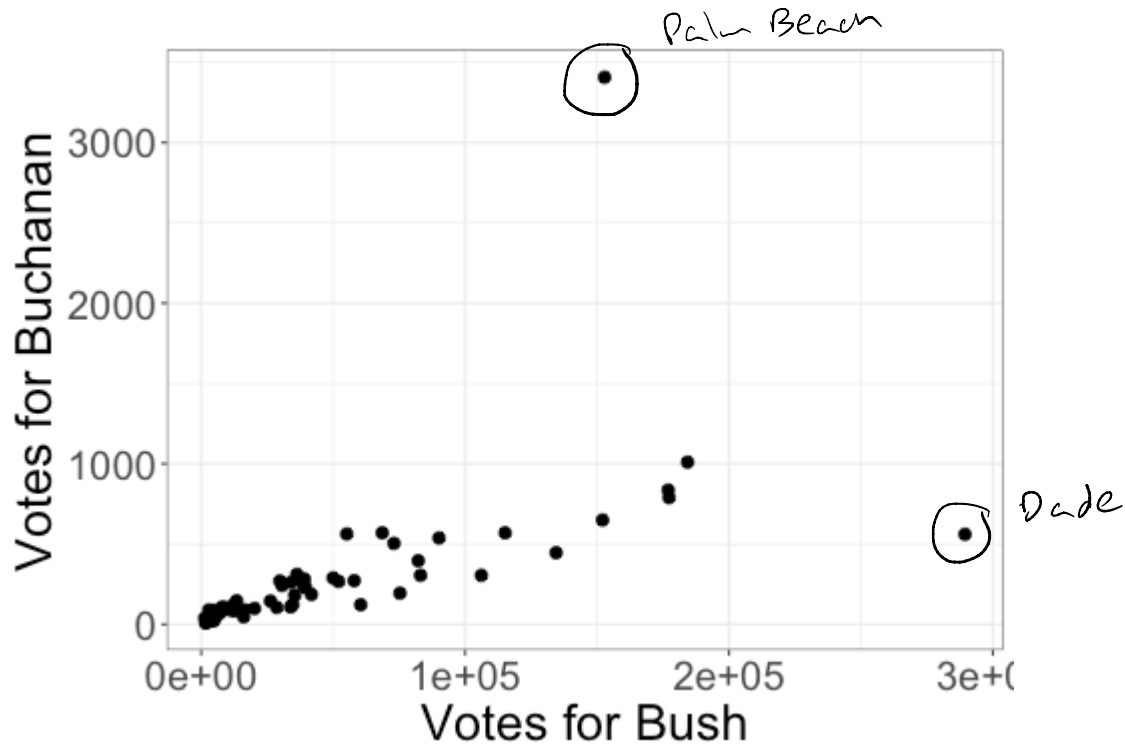


Outliers and influential points

Anything unusual?

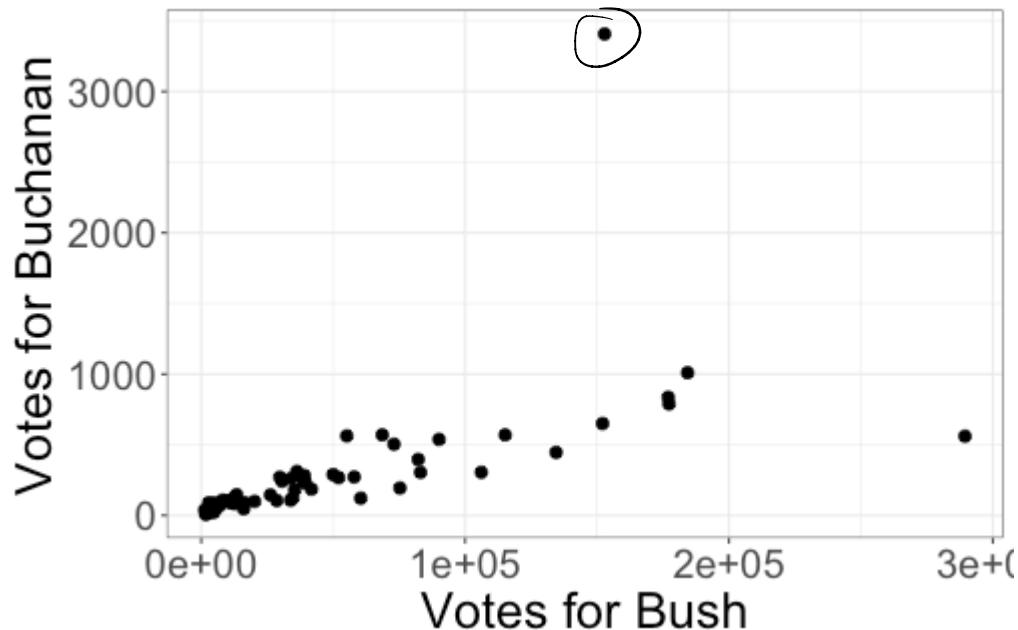
Votes for Bush and Buchanan, in Florida counties in 2000:



Does anything look unusual in this scatterplot?

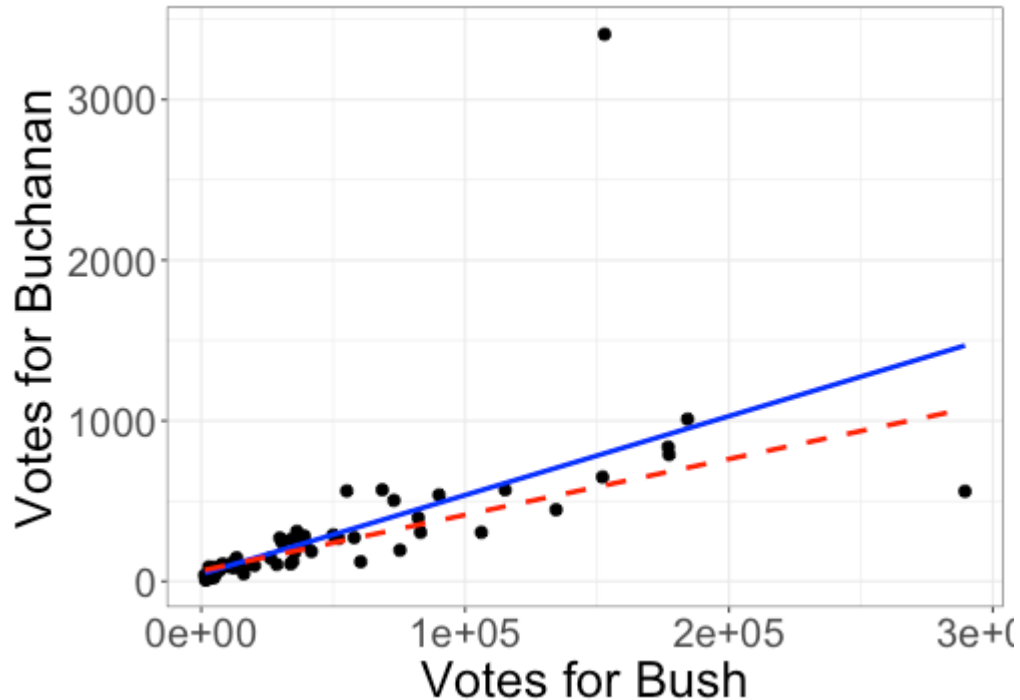
Anything unusual?

Outlier: A point which doesn't follow the general relationship between the predictor and response.



Why might the potential outlier be a problem?

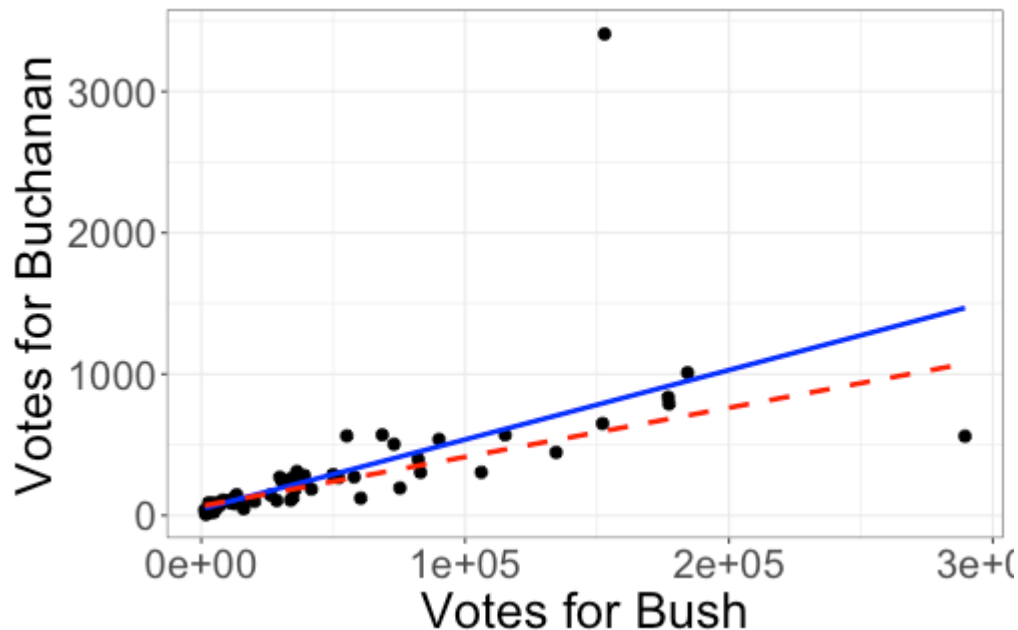
The impact of outliers



- + Blue, solid line: potential outlier is included in the data
- + Red, dashed line: potential outlier is excluded

Influential points

Outliers which substantially change the fitted line are called influential.

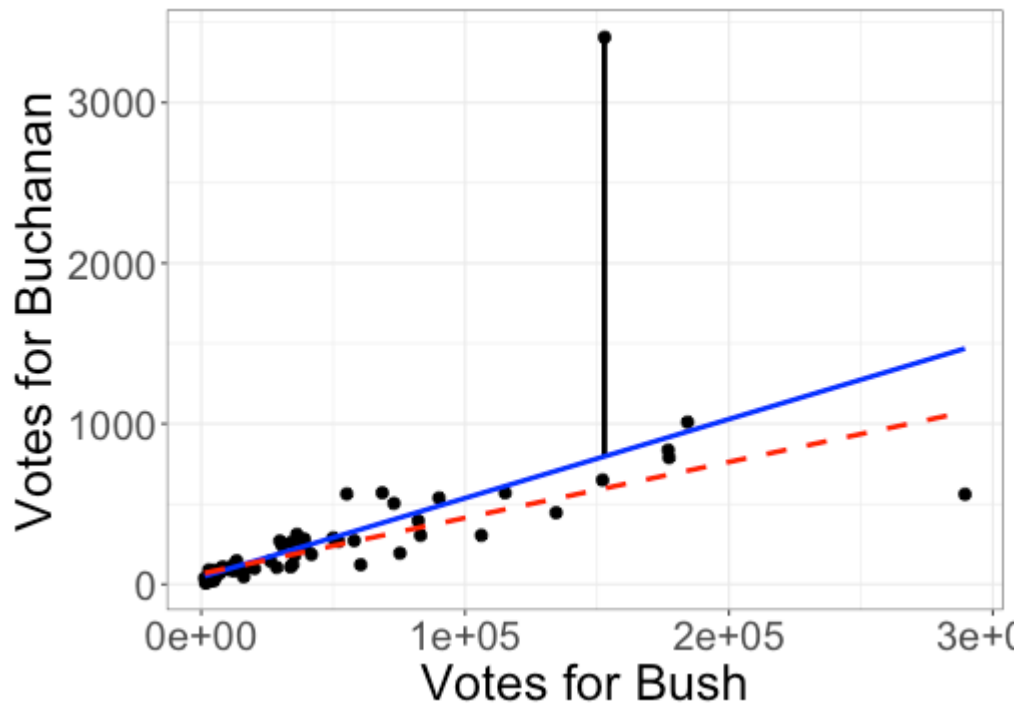


Why would an outlier change the fitted line?

Impact of influential points

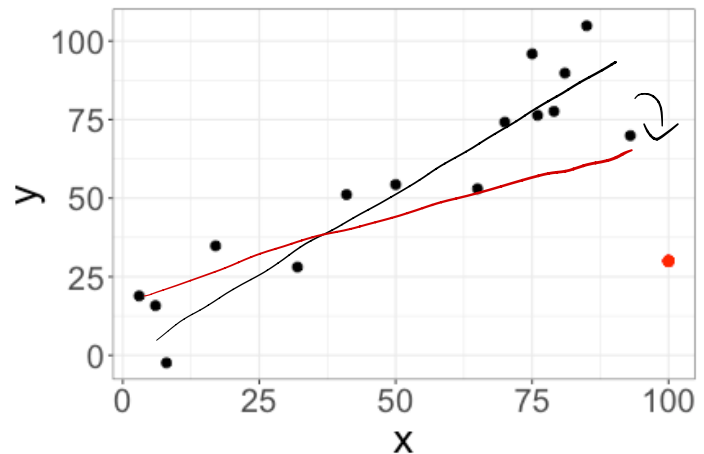
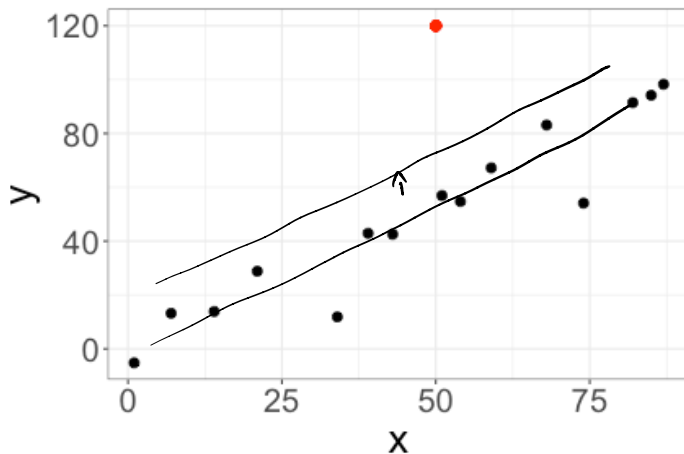
$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$$

$\hat{\beta}_0$ and $\hat{\beta}_1$ minimize $\sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2$ ↙ SS_E



Potential for influence

Which point do you think is more influential?



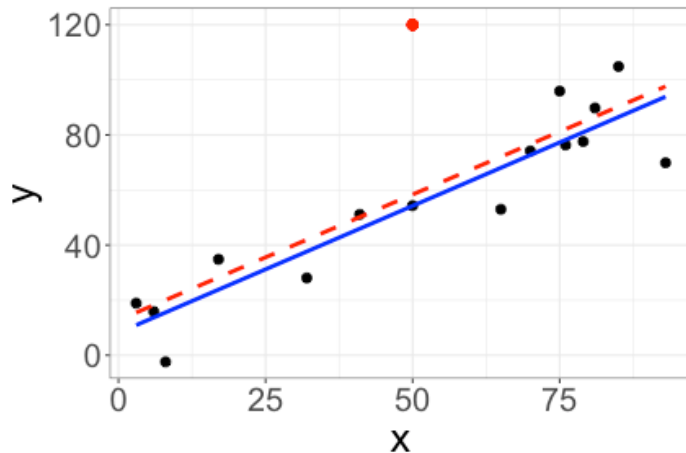
(A) The one on the left

(B) The one on the right

Potential for influence

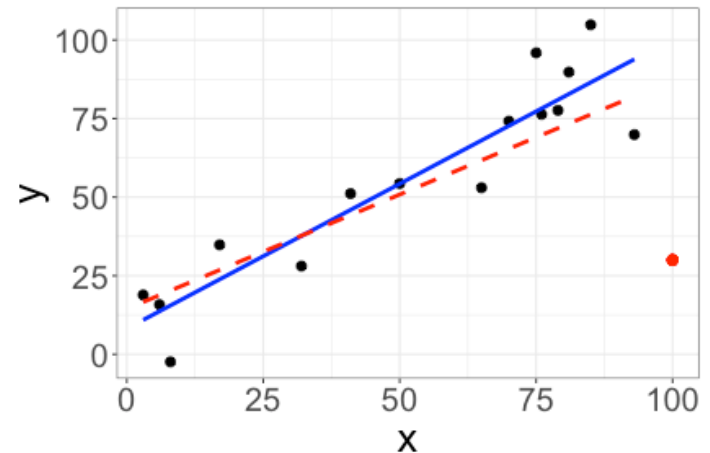
Influence depends on how far x is from $\bar{x}$.

Some influence



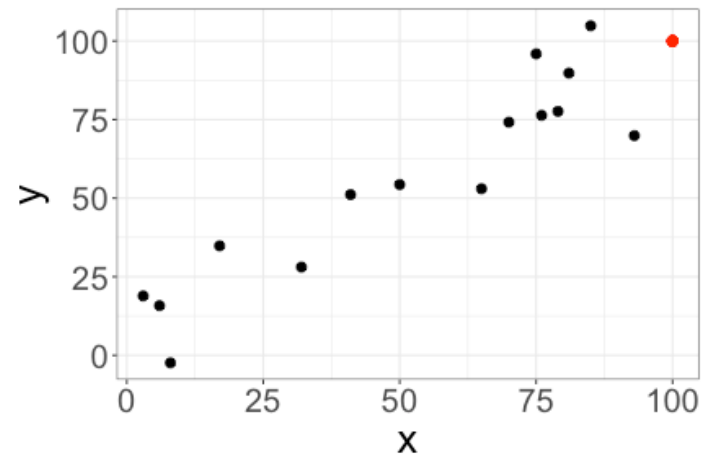
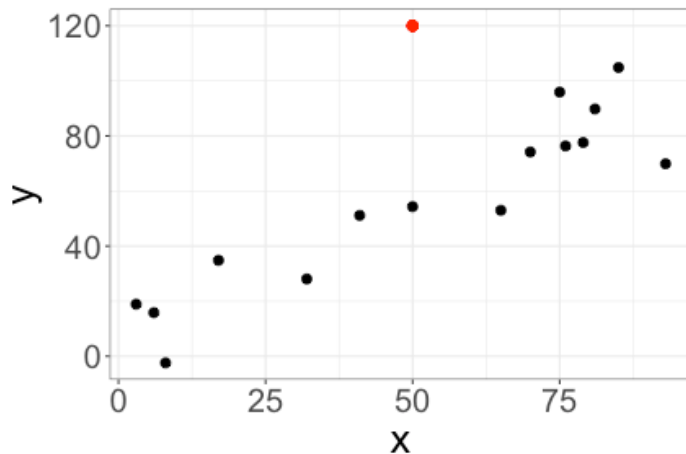
red: with red point
blue: without red point

More influence



Influential points

Which point do you think is more influential?



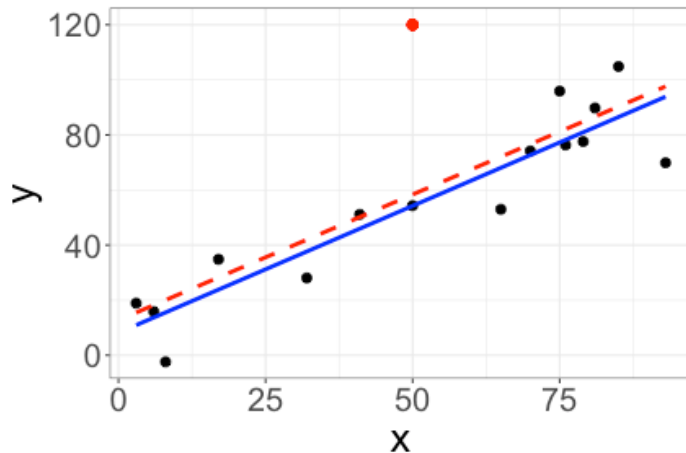
(A) The one on the left

(B) The one on the right

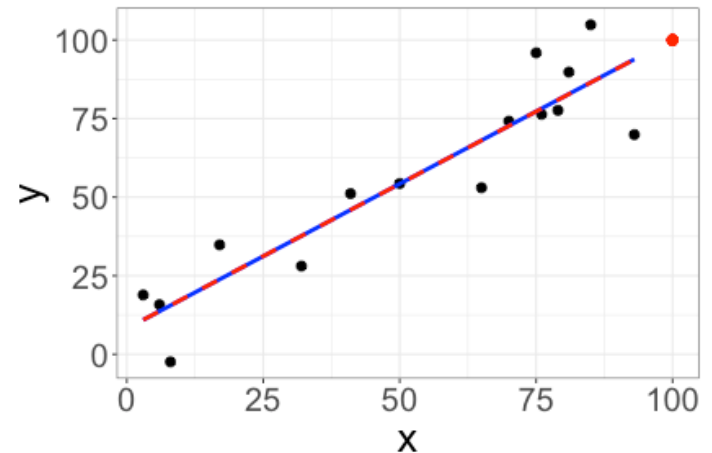
Influential points

Influence depends on both x and y .

Some influence



Not influential



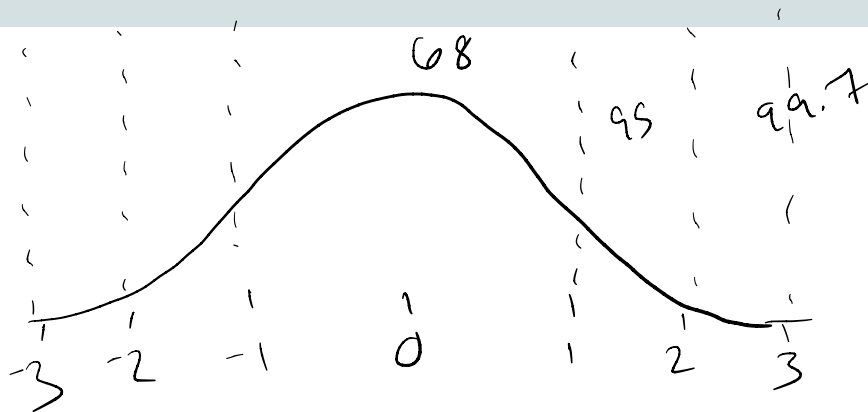
Assessing outliers: studentized residuals

Studentized residuals: $\frac{y_i - \hat{y}_i}{\hat{\sigma}_\varepsilon}$ $\leftarrow$ residual for i^{th} observation
 $\leftarrow$ standard deviation of all residuals (except i^{th} observation)

where $\hat{\sigma}_\varepsilon$ is computed without observation i .

- + Measure how many *standard deviations* away the observation is from the predicted value

What is a large number of standard deviations?

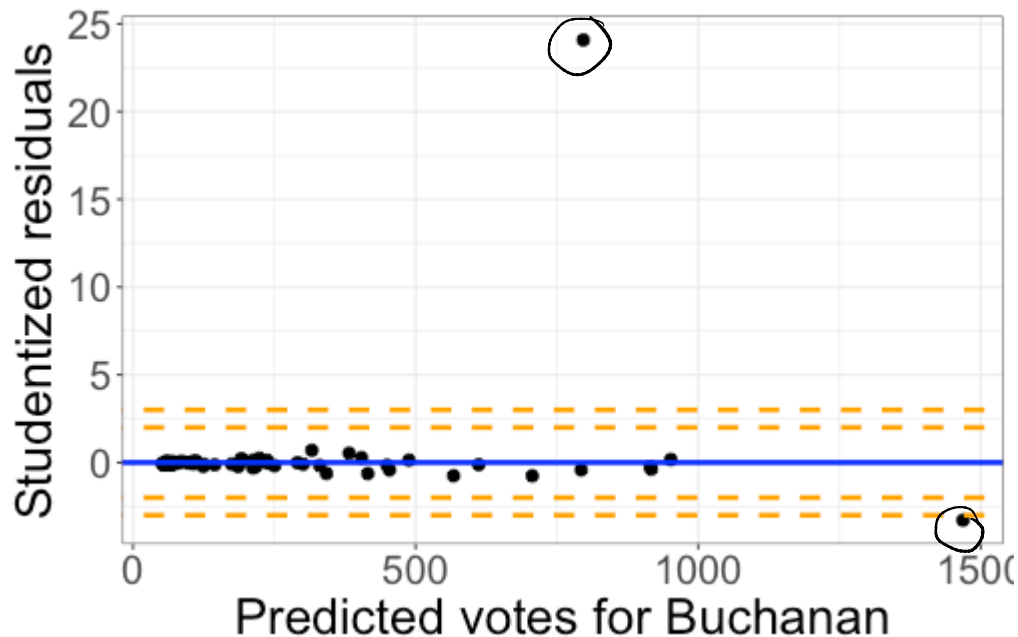


68% of data within $\pm 1SD$
95% of data within $\pm 2SD$
99.7% of data within $\pm 3SD$

Studentized residual plot

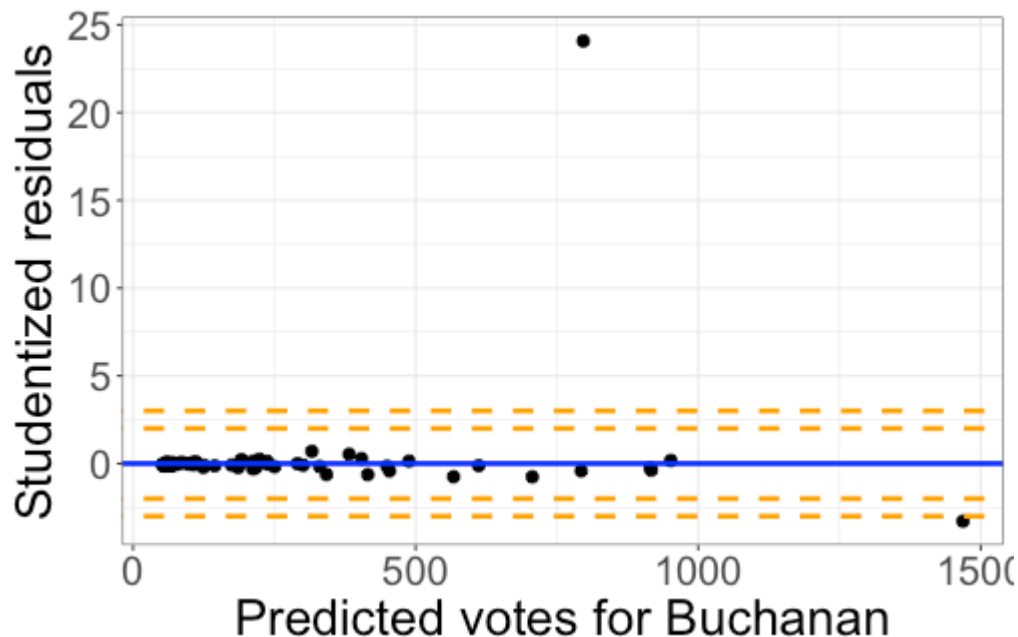
Studentized residuals: $\frac{y_i - \hat{y}_i}{\hat{\sigma}_\varepsilon}$

Studentized residual plot: plot studentized residuals on vertical axis, and $\hat{y}$ on horizontal axis. Add lines at -3, -2, 2, and 3.



Studentized residual plot

Studentized residual plot: plot studentized residuals on vertical axis, and $\hat{y}$ on horizontal axis. Add lines at -3, -2, 2, and 3.



Points < -3 or > 3 are suspected outliers.

Handling outliers

Suppose we identify a suspected outlier. What should we do about it?

Handling outliers

Suppose we identify a suspected outlier. What should we do about it?

If the observation is clearly a measurement error:

- + Remove it

Otherwise:

- + Fit line with and without point, see how the estimated line changes
- + If the estimated line changes substantially, report both fits

Class activity

https://sta112-s26.github.io/class_activities/ca_11.html

- + Exploring outliers
- + Submit HTML at end of class